GLOBAL ACADEMY OF TECHNOLOGY
(Autonomous) Approved by AICTE, New Delhi, Affiliated to VTU
Ideal Homes Township, Rajarajeshwari Nagar, Bengaluru - 560098
Department of Computer Science and Engineering
+91 80 28603158(Ext - 300) hodcse@gat.ac.in www.gat.ac.in

Academic Year: 2025-26 (Even Semester)

Major Project Phase-I (CSEP23605)

Review 0 - Synopsis Submission

Date	03/03/2026	**Domain**	Cyber Security
Group Number	G40	**Project ID**	P52
Project Title	AI-Powered Cybersecurity Platform for Real-Time Intrusion Detection		

Problem Description:

AI-Driven Cybersecurity Platform : Cyber attacks are growing in scale and complexity, while traditional security tools struggle to detect evolving threats.

Manual monitoring is impractical due to massive data volumes. There is a need for intelligent, automated systems that can learn from attack patterns and provide proactive defense.

Objectives:

Develop an AI-based model for anomaly and intrusion detection.

Analyze malware and phishing patterns using machine learning.

Build a scalable framework for real-time cyber threat prediction.

Automate Threat Response and Alert Generation

Literature survey Completed So far and Gap Identified through the Literature Survey

Sl. #	Title of the paper and year	Methodology	Advantages	Disadvantages
1	Dr. Docker (2025)	Uses DITector framework to analyze Docker images using dependency graphs and threat detection tools	Detects multiple threats (malware, secrets, vulnerabilities)	Depends on existing tools accuracy
2	CNN-LSTM Intrusion Detection (2025)	Combines CNN (feature extraction) + LSTM (sequence learning)	Detects complex attack patterns	High computational cost
3	ML-Based IDS using Random Forest (2024)	Uses supervised ML (Random Forest) for classification	Good for structured data	Cannot detect unknown attacks well

4	Deep Learning Cybersecurity Review (2024)	Survey of CNN, RNN, and hybrid models	Provides complete overview	No practical implementation
5	Autoencoder-Based Anomaly Detection (2025)	Uses unsupervised learning (autoencoders)	Detects zero-day attacks	High false positives
6	AI-Based Cloud Intrusion Detection (2024)	ML models for analyzing cloud traffic	Scalable	Needs continuous training
7	Microservices & Docker Security (2024)	Uses ML + monitoring tools for container security	Handles distributed systems	Complex deployment
8	Industrial ML Security System (2024)	Uses XGBoost for anomaly detection in industrial data	High performance	Requires domain-specific data
9	Explainable AI in Cybersecurity (2023)	Combines ML with explainable AI (XAI)	Transparent decision-making	Increased complexity
10	Generative AI in Cybersecurity (2024)	Uses LLMs for threat detection and analysis	Advanced threat detection	Risk of misuse

Panel Members' Remarks (if any):	
Status:	○ **Approved** ○ **Rejected** ○ **Revision needed**

	Name	**Signature**
Panel Member 1	Prof.B S Umashankar	
Panel Member 2	Prof.Ravi Babu G	
Project Guide	Prof.Sowmya B K	
Project Coordinator	Prof . Vanishree M L	
HOD, CSE	Dr. R Rajkumar Rao	